
Week 40: More on continuous-time Markov Chains

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

1. Repetition from last week

Theorem 1.1. (Forward Kolmogorov differential equations) For a birth-and-death process on $\mathcal{S} = \{0, 1, 2, \dots\}$,

$$\begin{aligned} P'_{i,0}(t) &= -\lambda_0 P_{i,0}(t) + \mu_1 P_{i,1}(t), \quad t \geq 0 \\ P'_{i,j}(t) &= \lambda_{j-1} P_{i,j-1}(t) - (\lambda_j + \mu_j) P_{i,j}(t) + \mu_{j+1} P_{i,j+1}(t), \quad t \geq 0, \quad j \geq 1, \end{aligned}$$

with initial conditions $P_{i,j}(0) = \delta_{i,j}$.

2. Theory

Theorem 2.1. For a birth-and-death process without absorbing states,

$$\pi_j = \lim_{t \rightarrow \infty} P_{i,j}(t), \quad j = 0, 1, \dots,$$

exists and are not dependent on i .

Remark:

- $\pi_j = 0$ for all states j is possible.
- If $\sum_{j=0}^{\infty} \pi_j = 1$, the $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ is called the limiting distribution.
- If the limiting distribution exists, it is also a stationary distribution. We can see this since

$$\pi_j = \lim_{s \rightarrow \infty} P_{i,j}(s+t) = \lim_{s \rightarrow \infty} \sum_{k=0}^{\infty} P_{i,k}(s) P_{k,j}(t) = \sum_{k=0}^{\infty} \pi_k P_{k,j}(t), \quad j = 0, 1, \dots, \quad t \geq 0.$$

The general importance of birth and death processes as models derives in large part from the availability of standard formulas for determining if a limiting distribution exists and what its values are when it does. These formulas follow from the Kolmogorov forward differential equations in [Theorem 1.1](#). Taking the limit as $t \rightarrow \infty$ in these equations, we see

$$\begin{aligned} 0 &= -\lambda_0 \pi_0 + \mu_1 \pi_1, \\ 0 &= \lambda_{j-1} \pi_{j-1} - (\lambda_j + \mu_j) \pi_j + \mu_{j+1} \pi_{j+1}, \quad j \geq 1. \end{aligned}$$

Here we get zeroes on the left-hand side because the limiting distribution is converging to a constant.

Theorem 2.2. When the limiting distribution $\boldsymbol{\pi}$ exists, it is the unique solution to

$$\begin{aligned} \lambda_0 \pi_0 &= \mu_1 \pi_1, \\ (\lambda_j + \mu_j) \pi_j &= \lambda_{j-1} \pi_{j-1} + \mu_{j+1} \pi_{j+1}, \quad j \geq 1, \\ \sum_{j=0}^{\infty} \pi_j &= 1. \end{aligned}$$

Remark: This is quite intuitive, as “Rate in = Rate out” with “Rate” being the number of events per unit time.

Theorem 2.3. For a birth-and-death process without absorbing states, we have

$$\pi_i = \theta_i \pi_0 = \frac{\theta_i}{\sum_{k=0}^{\infty} \theta_k}, \quad i = 0, 1, \dots,$$

where $\theta_0 = 1$ and

$$\theta_i = \frac{\lambda_0}{\mu_1} \cdot \frac{\lambda_1}{\mu_2} \cdot \dots \cdot \frac{\lambda_{i-1}}{\mu_i} = \frac{\prod_{j=0}^{i-1} \lambda_j}{\prod_{j=1}^i \mu_j}, \quad i = 1, 2, \dots$$

Theorem 2.4. For a birth-and-death process $\{X(t) : t \geq 0\}$ with state space $\mathcal{S}$, let $A \subset \mathcal{S}$ and let

$$v_i = \mathbb{E}[\min\{t \geq 0 : X(t) \in A\} \mid X(0) = i], \quad i \in \mathcal{S}.$$

If there are no absorbing states outside A , the expected times can be found by solving

$$\begin{aligned} v_i &= 0, \quad i \in A, \\ v_i &= \mathbb{E}[S_i] + \sum_{j \neq i} P(i \rightarrow j) v_j = \frac{1}{\lambda_i + \mu_i} + \sum_{j \neq i} P(i \rightarrow j) v_j, \quad i \notin A. \end{aligned}$$

Theorem 2.5. For a birth-and-death process, absorption probabilities are computed using the discrete-time Markov chain with one-step transition probabilities given by

$$P_{i,j} = P(i \rightarrow j), \quad i \neq j,$$

and

$$P_{i,i} = \begin{cases} 0 & \text{if } \lambda_i + \mu_i > 0 \\ 1 & \text{if } \lambda_i + \mu_i = 0 \end{cases}, \quad i = 0, 1, \dots$$

3. Problems

Problem 3.1. A machine works (1) or is broken (0). Time to breakdown is $\text{Exp}(\mu)$, and time to repair is $\text{Exp}(\lambda)$. All sojourn times are independent. The machine works at time $t = 0$. Calculate the probability that the machine works at time $t = 10$.

Solution. We have

$$X(t) = \begin{cases} 1 & \text{machine works at time } t \\ 0 & \text{machine broken at time } t \end{cases}$$

and the goal is to find $P_{1,1}(10) = P(X(10) = 1 \mid X(0) = 1)$. By the forward Kolmogorov differential equations, we have

$$P'_{1,1}(t) = \lambda P_{1,0}(t) - \mu P_{1,1}(t),$$

and initial conditions $P_{1,1}(0) = 1$. As we only have two states, we also know that $P_{1,0}(t) = 1 - P_{1,1}(t)$, and we can rewrite the differential equation as

$$P'_{1,1}(t) + \mu P_{1,1}(t) = \lambda [1 - P_{1,1}(t)] \iff P'_{1,1}(t) + (\lambda + \mu) P_{1,1}(t) = \lambda$$

Again, applying the integrating factor $e^{(\lambda+\mu)t}$ gives

$$\begin{aligned} e^{(\lambda+\mu)t} \left(\frac{d}{dt} P_{1,1}(t) + (\lambda + \mu) P_{1,1}(t) \right) &= \lambda e^{(\lambda+\mu)t} \\ \implies \frac{d}{dt} (e^{(\lambda+\mu)t} P_{1,1}(t)) &= \lambda e^{(\lambda+\mu)t} \\ \implies e^{(\lambda+\mu)t} P_{1,1}(t) &= \int_0^t \lambda e^{(\lambda+\mu)\tau} d\tau + C = \frac{\lambda}{\lambda + \mu} (e^{(\lambda+\mu)t} - 1) + C \\ \implies P_{1,1}(t) &= \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda+\mu)t}) + C e^{-(\lambda+\mu)t}. \end{aligned}$$

The initial value $P_{1,1}(0) = 1$ gives $C = 1$, meaning

$$P_{1,1}(10) = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda+\mu)10}) + e^{-(\lambda+\mu)10} = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda+\mu)10}.$$

Problem 3.2. (Problem 1, Exercise 6) A pure birth process starting from $X(0) = 0$ has birth parameters $\lambda_0 = 1$, $\lambda_1 = 3$, $\lambda_2 = 2$ and $\lambda_3 = 5$. Let W_1 , W_2 and W_3 be the stochastic times it takes the process to reach states 1, 2 and 3, respectively.

1. Determine the transition probability functions $P_n(t) = P(X(t) = n | X(0) = 0)$ for $n = 0, 1, 2, 3$.
2. Write W_3 as a sum of sojourn times and thereby deduce that the mean time is $\mathbb{E}[W_3] = 11/6$.
3. Determine the mean of $W_1 + W_2 + W_3$.
4. Determine the variance of W_3 .

Solution.

1. By the forward Kolmogorov differential equations, we have

$$\begin{aligned} P'_0(t) &= -\lambda_0 P_0(t), \\ P'_n(t) &= \lambda_{n-1} P_{n-1}(t) - \lambda_n P_n(t), \quad n = 1, 2, 3 \end{aligned}$$

where the initial conditions are $P_0(0) = 1$ and $P_n(0) = 0$ for $n = 1, 2, 3$.

We have to solve the system of differential equations recursively. Starting with $P_0(t)$, we have the integrating factor $e^{\int_0^t \lambda_0 d\tau} = e^{\lambda_0 t}$, so

$$\begin{aligned} e^{\lambda_0 t} \left(\frac{d}{dt} P_0(t) + \lambda_0 P_0(t) \right) &= \frac{d}{dt} (e^{\lambda_0 t} P_0(t)) = 0 \\ \implies e^{\lambda_0 t} P_0(t) &= C \\ \implies P_0(t) &= C e^{-\lambda_0 t}. \end{aligned}$$

With the initial condition $P_0(0) = 1$, we find $C = 1$, giving $P_0(t) = e^{-\lambda_0 t} = e^{-t}$.

For state 1, we have

$$P'_1(t) + \lambda_1 P_1(t) = \lambda_0 P_0(t) = e^{-t}.$$

Using the integrating factor $e^{\int_0^t \lambda_1 d\tau} = e^{\lambda_1 t}$, we have

$$\begin{aligned} e^{\lambda_1 t} \left(\frac{d}{dt} P_1(t) + \lambda_1 P_1(t) \right) &= \frac{d}{dt} (e^{\lambda_1 t} P_1(t)) = e^{\lambda_1 t} e^{-t} = e^{(\lambda_1 - 1)t} \\ \implies e^{\lambda_1 t} P_1(t) &= \int_0^t e^{(\lambda_1 - 1)\tau} d\tau + C = \frac{1}{\lambda_1 - 1} (e^{(\lambda_1 - 1)t} - 1) + C \\ \implies P_1(t) &= \frac{1}{\lambda_1 - 1} (e^{-t} - e^{-\lambda_1 t}) + C e^{-\lambda_1 t}. \end{aligned}$$

and using the initial condition $P_1(0) = 0$, we find $C = 0$, giving

$$P_1(t) = \frac{1}{\lambda_1 - 1} (e^{-t} - e^{-\lambda_1 t}) = \frac{1}{2} (e^{-t} - e^{-3t}).$$

The same method can be applied to the remaining equations, giving

$$\begin{aligned} P_2(t) &= \frac{3}{2} (e^{-t} - 2e^{-2t} + e^{-3t}), \\ P_3(t) &= \frac{3}{4} e^{-t} - 2e^{-2t} + \frac{3}{2} e^{-3t} - \frac{1}{4} e^{-5t}. \end{aligned}$$

2. Letting S_0 , S_1 and S_2 be the sojourn times, we want to estimate $W_3 = S_0 + S_1 + S_2$. As $S_i \sim \text{Exp}(\lambda_i)$ and using the linearity of the expectation

$$\mathbb{E}[W_3] = \sum_{i=0}^2 \mathbb{E}[S_i] = \frac{1}{\lambda_0} + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} = 1 + \frac{1}{3} + \frac{1}{2} = \frac{11}{6}.$$

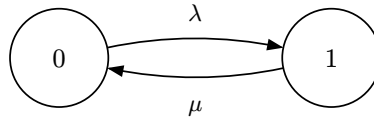
3. Since $W_1 = S_0$, $W_2 = S_0 + S_1$, and $W_3 = S_0 + S_1 + S_2$, we have

$$\mathbb{E}[W_1 + W_2 + W_3] = 3\mathbb{E}[S_0] + 2\mathbb{E}[S_1] + \mathbb{E}[S_2] = 3 + \frac{2}{3} + \frac{1}{2} = \frac{25}{6}.$$

4. Since the sojourn times in this pure birth process are independent, we have

$$\text{Var}[W_3] = \sum_{i=0}^2 \text{Var}[S_i] = \frac{1}{\lambda_0^2} + \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2} = 1 + \frac{1}{9} + \frac{1}{4} = \frac{49}{36}.$$

Problem 3.3. Consider a CTMC



with $\lambda, \mu > 0$. Derive the long-run proportion of time spent in state 0.

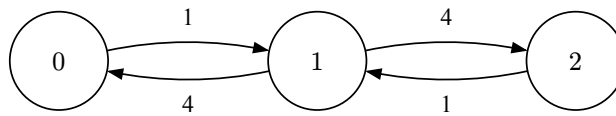
Solution. Recall that “Rate in = Rate out”. Thus

$$\pi_0 \lambda = \pi_1 \mu \implies \pi_1 = \frac{\lambda}{\mu} \pi_0,$$

giving

$$\pi_0 + \pi_1 = \pi_0 \left(1 + \frac{\lambda}{\mu} \right) = 1 \implies \pi_0 = \frac{\mu}{\lambda + \mu}.$$

Problem 3.4. Consider a CTMC



Calculate π_0 , π_1 and π_2 .

Solution. See

$$\theta_0 = 1, \quad \theta_1 = \frac{\lambda_0}{\mu_1} = \frac{1}{4}, \quad \theta_2 = \theta_1 \cdot \frac{\lambda_1}{\mu_2} = \frac{1}{4} \cdot \frac{4}{1} = 1,$$

so

$$\sum_{k=0}^2 \theta_k = 1 + \frac{1}{4} + 1 = \frac{9}{4}.$$

Giving

$$\pi_0 = \frac{1}{\sum_{k=0}^2 \theta_k} = \frac{4}{9}, \quad \pi_1 = \theta_1 \pi_0 = \frac{1}{4} \cdot \frac{4}{9} = \frac{1}{9}, \quad \pi_2 = \theta_2 \pi_0 = 1 \cdot \frac{4}{9} = \frac{4}{9}.$$

Problem 3.5. Customers arrive in a store according to a Poisson process with rate $\lambda = 60$ customers per hour. The store has one server, and all customers in the service area are either in service or waiting in a queue. An arrival begins service immediately if the server is idle. If the server is busy, the customer joins the queue with probability 1 when there are at most two customers in the service area. With three or more, the customer joins the queue with probability $0 < p < 1$, or leaves immediately with probability $1 - p$. Joining decisions are independent across customers. Service times are i.i.d. exponential with mean $1/\mu = 6$ minutes and are independent of the arrival process. Whenever service is completed, the customer at the front of the queue immediately begins service.

Let $X(t)$ denote the number of customers either queuing or receiving service at time t .

1. Assume that time is measured in hours, and determine the birth rates and the death rates as functions of p . Draw the transition diagram.
2. Let $p = 0.1$, and assume that there are currently three customers queuing or receiving service.
 - Calculate the probability that exactly two customers finish their service times before the next customer joins the queue.
 - Calculate the expected time until the next customer joins the queue.
3. Derive the limiting probabilities $\pi_i = \lim_{t \rightarrow \infty} P(X(t) = i)$ for $i = 0, 1, \dots$ as functions of p .
4. What is the necessary condition on p for $\{X(t) : t \geq 0\}$ to have a limiting distribution? Give an intuitive explanation for the condition.

Solution.

1. The birth rates are

$$\lambda_0 = \lambda_1 = \lambda_2 = \lambda = 60, \quad \lambda_i = 60p, \quad i \geq 3,$$

while the death rates converted to hours are $\mu = 10$.

2. • We need jumps $3 \rightarrow 2 \rightarrow 1 \rightarrow 2$ for **exactly** two customers finish before a new customer joins. Computing using the jump probabilities, we have

$$\begin{aligned} P(3 \rightarrow 2 \rightarrow 1 \rightarrow 2) &= P(3 \rightarrow 2)P(2 \rightarrow 1)P(1 \rightarrow 2) \\ &= \left(\frac{\mu_3}{\lambda_3 + \mu_3} \right) \left(\frac{\mu_2}{\lambda_2 + \mu_2} \right) \left(\frac{\lambda_1}{\lambda_1 + \mu_1} \right) \\ &= \left(\frac{10}{6 + 10} \right) \left(\frac{10}{60 + 10} \right) \left(\frac{60}{60 + 10} \right) \approx 0.0765. \end{aligned}$$

- Let

$$v_i = \mathbb{E}[\text{minimum time until next customer joins queue} \mid X(0) = i], \quad i = 0, 1, \dots$$

Clearly, $v_0 = v_1 = v_2 = 1/\lambda = 1/60$. We are interested in v_3 . A first step analysis gives

$$\begin{aligned} v_3 &= \mathbb{E}[S_3] + P(3 \rightarrow 2)v_2 + P(3 \rightarrow 4) \cdot 0 \\ &= \frac{1}{\lambda_3 + \mu_3} + \left(\frac{\mu_3}{\lambda_3 + \mu_3} \right) \cdot v_2 \\ &= \frac{1}{6 + 10} + \frac{10}{6 + 10} \cdot \frac{1}{60} \approx 0.0729. \end{aligned}$$

3. From the formulas we have

$$\theta_k = \begin{cases} 1 & k = 0 \\ \frac{\lambda_0 \cdots \lambda_{k-1}}{\mu_1 \cdots \mu_k} & k \geq 1 \end{cases} \quad \text{and} \quad \pi_k = \begin{cases} \frac{1}{\sum_{i=0}^{\infty} \theta_i} & k = 0 \\ \pi_0 \theta_k & k \geq 1 \end{cases}.$$

We see

$$\theta_k = \begin{cases} \left(\frac{\lambda}{\mu}\right)^k & k \leq 2 \\ \left(\frac{\lambda}{\mu}\right)^3 \left(\frac{\lambda p}{\mu}\right)^{k-3} & k \geq 3 \end{cases}.$$

Considering the sum of θ_k , we have

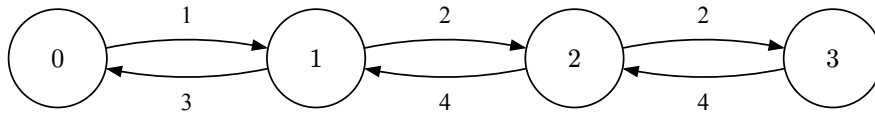
$$\begin{aligned} \sum_{k=0}^{\infty} \theta_k &= \sum_{k=0}^2 \left(\frac{\lambda}{\mu}\right)^k + \left(\frac{\lambda}{\mu}\right)^3 \sum_{k=0}^{\infty} \left(\frac{\lambda p}{\mu}\right)^k \\ &= \begin{cases} \infty & p \geq \frac{\mu}{\lambda} \\ 1 + \left(\frac{\lambda}{\mu}\right) + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 \frac{\mu}{\mu - \lambda p} & p < \frac{\mu}{\lambda} \end{cases} \end{aligned}$$

so

$$\pi_0 = \begin{cases} 0 & p \geq \frac{\mu}{\lambda} \\ \left(1 + \left(\frac{\lambda}{\mu}\right) + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 \frac{\mu}{\mu - \lambda p}\right)^{-1} & p < \frac{\mu}{\lambda} \end{cases}.$$

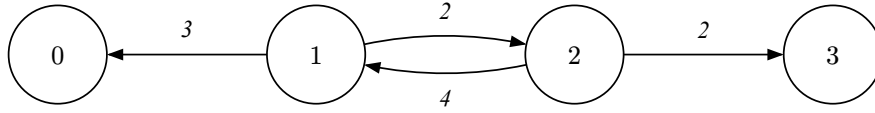
4. See that if $p \geq \mu/\lambda$, the $\pi_0 = 0$ and therefore also $\pi_k = \theta_k \pi_0 = 0$. This means that the limiting distribution does not exist. For $p < \mu/\lambda = 10/60 = 1/6$, the limiting distribution exists. The intuition is that we have to be able to process customers faster than they arrive, which can be expressed as the death rate being larger than the birth rate $\mu > \lambda p$.

Problem 3.6. Consider a CTMC



Let $X(0) = 1$, compute the probability that we reach state 3 before state 0.

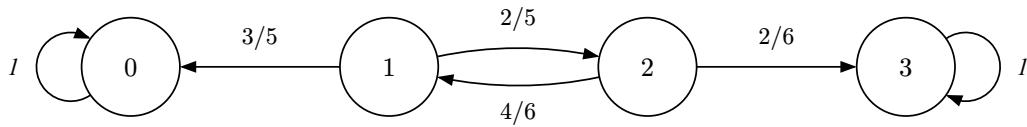
Solution. First we make states 0 and 3 absorbing, giving



Moving from a CTMC to a DTMC $\{Y_n : n = 0, 1, \dots\}$ with transition probability matrix $\mathbf{P}$. By [Theorem 2.5](#), we have

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{3}{5} & 0 & \frac{2}{5} & 0 \\ 0 & \frac{4}{6} & 0 & \frac{2}{6} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

and the diagram



Letting

$$u_i = P\{\text{absorption in 3} \mid Y_0 = i\}, \quad i = 0, 1, 2, 3,$$

we by first-step analysis see

$$u_0 = 0,$$

$$u_3 = 1,$$

$$u_1 = P_{1,0}u_0 + P_{1,2}u_2 = \frac{2}{5}u_2,$$

$$u_2 = P_{2,1}u_1 + P_{2,3}u_3 = \frac{2}{3}u_1 + \frac{1}{3}.$$

Solving this system we find $u_1 = 2/11$ (and $u_2 = 5/11$).